

# LEARNING MANAGEMENT SYSTEM (LMS)

## Client Requirements & Mini Project Brief | MySQL + Python + Streamlit

| Client | Learning & Training Institute                   | Project Type | Database + Web Application            |
|--------|-------------------------------------------------|--------------|---------------------------------------|
| Users  | Administrator, Academic Coordinator, Instructor | Technology   | MySQL, SQL, Python, Streamlit, Pandas |

### 1. Client Requirement

The institute needs a centralized LMS to replace spreadsheet-based management of students, instructors, courses, enrollments and academic results. The solution must provide a simple web interface, reliable relational data management, CRUD operations and management reports.

### 2. Business Objectives

- Centralize academic and administrative records.
- Manage students, instructors and courses from one system.
- Enroll students into courses and track enrollment status.
- Record student marks and review performance.
- Provide live KPIs and useful management reports.
- Demonstrate a maintainable MySQL + Python + Streamlit solution.

### 3. Functional Requirements

| Module      | Required Functionality                                                                   |
|-------------|------------------------------------------------------------------------------------------|
| Students    | Add, view, search, update and delete; maintain email, phone, enrollment date and status. |
| Instructors | Add, view, update and delete; maintain specialization, email and status.                 |
| Courses     | Create and manage course name, description, instructor, duration, fee and status.        |
| Enrollments | Enroll a student in a course; view history; update Enrolled/Completed/Dropped status.    |
| Grades      | Record assignment, quiz and final-exam marks; calculate total and review performance.    |
| Dashboard   | Show student, instructor, course and enrollment KPIs plus basic charts/tables.           |
| Reports     | Course popularity, instructor workload, student performance and enrollment status.       |

### 4. Database Requirements

| Table       | Purpose / Relationship                                          |
|-------------|-----------------------------------------------------------------|
| students    | Student master data; one student → many enrollments.            |
| instructors | Instructor master data; one instructor → many courses.          |
| courses     | Course master data; references instructor.                      |
| enrollments | Links students and courses; resolves many-to-many relationship. |
| grades      | Marks associated with an enrollment.                            |

Database rules: use Primary Keys, Foreign Keys, NOT NULL, UNIQUE email, appropriate data types, controlled status values and a UNIQUE(student\_id, course\_id) rule for duplicate enrollment prevention.

### 5. CRUD & SQL Skills

Students must demonstrate: CREATE TABLE, INSERT, SELECT, UPDATE, DELETE, WHERE, LIKE, ORDER BY, GROUP BY, COUNT, SUM, INNER/LEFT JOIN, aliases, calculated fields, primary keys, foreign keys and constraints.

### 6. Streamlit Application

Navigation: Dashboard | Students | Instructors | Courses | Enrollments | Grades | Reports. Use forms, dropdowns for related records, searchable tables, KPI cards, validation, success/error messages and confirmation for destructive actions. Python must connect to MySQL using parameterized queries.

## 7. Key Business Questions / Reports

| ID    | Client Question                                             | SQL Concept                       |
|-------|-------------------------------------------------------------|-----------------------------------|
| BR-01 | Which courses have the highest number of enrolled students? | JOIN + COUNT + GROUP BY           |
| BR-02 | How many courses is each instructor teaching?               | LEFT JOIN + COUNT                 |
| BR-03 | Which students have the highest total marks?                | JOIN + SUM/Calculation + ORDER BY |
| BR-04 | How many students are active?                               | COUNT + WHERE                     |
| BR-05 | How many enrollments are Enrolled, Completed or Dropped?    | GROUP BY + COUNT                  |
| BR-06 | Which courses have no students enrolled?                    | LEFT JOIN + NULL filtering        |

## 8. Client Acceptance Criteria

- MySQL database can be created successfully from the submitted SQL script.
- All five core tables and relationships work correctly.
- CRUD is demonstrated for students, instructors, courses, enrollments and grades.
- Streamlit displays and updates live MySQL data.
- Duplicate/invalid data is prevented or handled with clear messages.
- At least one multi-table JOIN report works correctly.
- Dashboard KPIs and reports are based on live database data.
- Project includes setup instructions and can be run by another user.

## 9. Student / Developer Deliverables

| Deliverable   | Expected Output                                               |
|---------------|---------------------------------------------------------------|
| Database      | lms_database.sql with schema, constraints and sample data.    |
| Application   | app.py / Python modules with MySQL connection and CRUD logic. |
| UI            | Working Streamlit LMS screens and dashboard.                  |
| Reports       | Required SQL queries with displayed results.                  |
| Documentation | README: setup, configuration, execution and screenshots.      |
| Demo          | Short presentation demonstrating end-to-end workflow.         |

## 10. Suggested Project Workflow

Database Design → MySQL Schema → Sample Data → SQL CRUD → Python-MySQL Connection → Streamlit CRUD → Dashboard → Reports → Testing & Demo

## 11. Assessment Rubric

| Area                                     | Marks |
|------------------------------------------|-------|
| Database design, keys & relationships    | 15    |
| SQL CRUD & JOIN/aggregation              | 25    |
| Python-MySQL integration                 | 10    |
| Streamlit CRUD application               | 20    |
| Dashboard & reports                      | 15    |
| Validation, documentation & presentation | 15    |
| Total                                    | 100   |

## 12. Optional Extensions

Attendance | Fees/Payments | Certificates | User Login & Roles | Batches | Course Completion % | CSV/Excel Export | Stored Procedures | Advanced Analytics